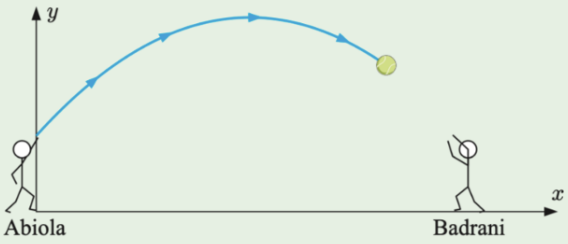
Chapter 3 : Quadratics

Mme Doux

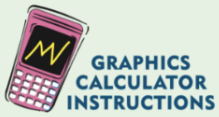
I - Quadratic equations

Investigation

Abiola and Badrani are standing 40 metres apart, throwing a ball between them. When Abiola throws the ball, it travels in a smooth arc. At the time when the ball has travelled x metres horizontally towards Badrani, its height is y metres.







x (m)	0	5	10	15	20	25	30
y (m)	1.25	10	16.25	20	21.25	20	16.25

Things to think about:

a Use technology to plot these points.

b What *shape* is the graph of y against x?

c What is the maximum height reached by the ball?

d What *formula* gives the height of the ball when it has travelled x metres horizontally towards Badrani?

e Will the ball reach Badrani before it bounces?

Definition

A quadratic equation is an equation of the form

$$ax^2 + bx + c = 0 \quad \text{where } a, b, c \text{ are constants, } a \neq 0.$$

Definition

A quadratic function is a function of the form

$$y = ax^2 + bx + c, \quad a \neq 0.$$

Remark. Quadratic functions are members of the family of **polynomials**. The first few members of this family are shown in the table:

Polynomial function	Type
$y = ax + b,\ a \neq 0$	linear
$y = ax^2 + bx + c,\ a \neq 0$	quadratic
$y = ax^3 + bx^2 + cx + d,\ a \neq 0$	cubic
$y = ax^4 + bx^3 + cx^2 + dx + e,\ a \neq 0$	quartic

To solve quadratic equations we have the following methods to choose from:

- factorise the quadratic and use the **Null Factor law:**

$$\text{If } ab = 0 \text{ then } a = 0 \text{ or } b = 0.$$

- complete the square
- use the quadratic formula
- use technology

Definition

The roots or solutions of

$$ax^2 + bx + c = 0$$

are the values of x which satisfy the equation.

Example

Is 2 a root of $x^2 - 3x + 2$?

Example

Solve for x:

a) $3x^2 + 5x = 0$

b) $x^2 = 5x + 6$

Example

Solve for x:

a) $4x^2 + 1 = 4x$

b) $6x^2 = 11x + 10$

Example

Solve for x:

a) $(x + 2)^2 = 7$

b) $(x - 1)^2 = -5$

Example

Solve for exact values of x by completing the square:

$$x^2 + 4x + 1 = 0$$

II - Quadratic formula

Property

$$\text{If } ax^2 + bx + c = 0 \text{ and } \quad \Delta = b^2 - 4ac$$

- If $\Delta = 0$, $\quad x = \dfrac{-b}{2a}$ is **the only solution** (a repeated or double root).
- If $\Delta > 0$, $\quad \sqrt{\Delta}$ is a positive real number, so there are **two distinct real roots**:

$$x = \frac{-b + \sqrt{\Delta}}{2a} \quad \text{and} \quad x = \frac{-b - \sqrt{\Delta}}{2a}$$

- If $\Delta < 0$, $\quad \sqrt{\Delta}$ is not a real number and so there are **no real roots**.
- If a, b, c are rational and Δ is a **square** then the equation has two rational roots which can be found by factorisation.

Proof

$$ax^2 + bx + c = 0, \quad a \neq 0$$

$$\Rightarrow x^2 + \frac{b}{a}x + \frac{c}{a} = 0 \quad \{\text{dividing each term by } a\}$$

$$\Rightarrow x^2 + \frac{b}{a}x = -\frac{c}{a}$$

$$\Rightarrow x^2 + \frac{b}{a}x + \left(\frac{b}{2a}\right)^2 = -\frac{c}{a} + \left(\frac{b}{2a}\right)^2 \quad \{\text{completing the square}\}$$

$$\Rightarrow \left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2} \quad \{\text{factorising}\}$$

$$\Rightarrow x + \frac{b}{2a} = \pm\sqrt{\frac{b^2 - 4ac}{4a^2}}$$

$$\Rightarrow x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Example

Solve for x:

a) $x^2 - 2x - 6 = 0$

b) $2x^2 + 3x - 6 = 0$

Example

Consider the equation $kx^2 + (k+3)x - 1 = 0$. Find the discriminant Δ and draw its sign diagram. Hence, find the value of k for which the equation has:

1. two distinct real roots
2. two real roots
3. a repeated root
4. no real roots

III - Quadratic functions

Definition

A quadratic function has the form

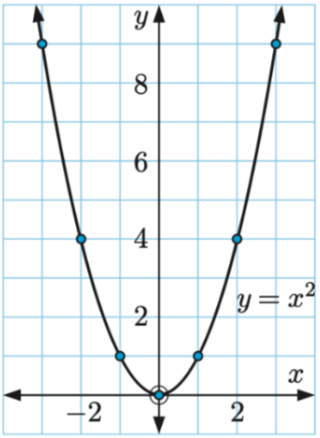
$$y = ax^2 + bx + c \quad \text{where } a \neq 0.$$

The simplest quadratic function is

$$y = x^2$$

Its graph can be drawn from a table of values:

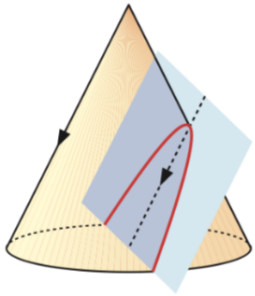
x	-3	-2	-1	0	1	2	3
y	9	4	1	0	1	4	9



The graph of a quadratic function is called a **parabola**.

The parabola is one of the **conic sections**, the others being circles, hyperbolae and ellipses.

They are called conic sections because they can be obtained by cutting a cone with a plane.



A parabola is produced by cutting the cone with a plane parallel to its slant side.

Investigation

1. (a) Use technology to help you sketch:

$$y = (x-1)(x-3), \quad y = 2(x-1)(x-3), \quad y = -(x-1)(x-3),$$

$$y = -3(x-1)(x-3), \quad y = -\tfrac{1}{2}(x-1)(x-3)$$

   (b) Find the x-intercepts for each function in (a).

   (c) What is the geometrical significance of a in $y = a(x-1)(x-3)$?

2. (a) Use technology to help you sketch:

$$y = 2(x-1)(x-4), \quad y = 2(x-3)(x-5), \quad y = 2(x+1)(x-2),$$

$$y = 2x(x+5), \quad y = 2(x+2)(x+4)$$

   (b) Find the x-intercepts for each function in (a).

   (c) What is the geometrical significance of p and q in $y = 2(x-p)(x-q)$?

3. (a) Use technology to help you sketch:

$$y = 2(x-1)^2, \quad y = 2(x-3)^2, \quad y = 2(x+2)^2, \quad y = 2x^2$$

   (b) Find the x-intercepts for each function in (a).

   (c) What is the geometrical significance of p in $y = 2(x-p)^2$?

4. Copy and complete:

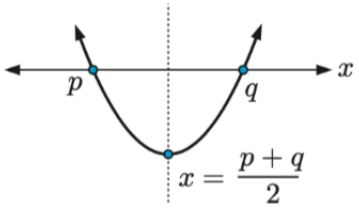
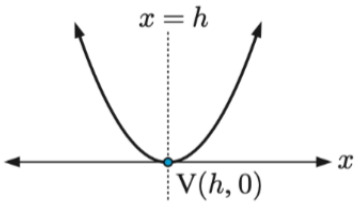
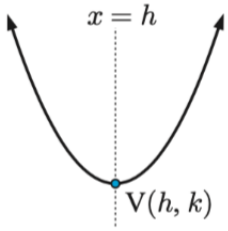
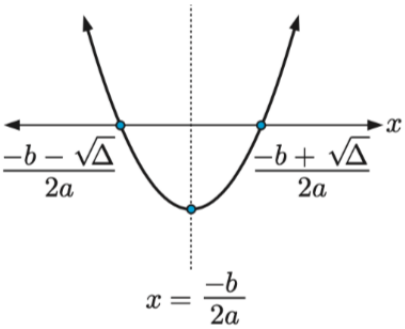
   - If a quadratic has the form $y = a(x-p)(x-q)$ then it ... the x-axis at ...
   - If a quadratic has the form $y = a(x-p)^2$ then it ... the x-axis at ...

Property

For a quadratic function

$$y = ax^2 + bx + c, \quad a \neq 0 :$$

- $a > 0$ produces the shape  called **concave up.**
- $a < 0$ produces the shape  called **concave down.**
- If $-1 < a < 1$, $a \neq 0$, the graph is **wider** than $y = x^2$.
- If $a < -1$ or $a > 1$, the graph is **narrower** than $y = x^2$.

Quadratic form, $a \neq 0$	*Graph*	*Facts*
• $y = a(x-p)(x-q)$ p, q are real		x-intercepts are p and q axis of symmetry is $x = \frac{p+q}{2}$ vertex is $\left(\frac{p+q}{2}, f\left(\frac{p+q}{2}\right)\right)$
• $y = a(x-h)^2$ h is real		touches x-axis at h axis of symmetry is $x = h$ vertex is $(h, 0)$
• $y = a(x-h)^2 + k$		axis of symmetry is $x = h$ vertex is (h, k)
• $y = ax^2 + bx + c$		y-intercept c axis of symmetry is $x = \frac{-b}{2a}$ vertex is $\left(-\frac{b}{2a}, c - \frac{b^2}{4a}\right)$ x-intercepts for $\Delta \geqslant 0$ are $\frac{-b \pm \sqrt{\Delta}}{2a}$ where $\Delta = b^2 - 4ac$

Example

Using axes intercepts only, sketch the graphs of:

1. $y = 2(x+3)(x-1)$
2. $y = -2(x-1)(x-2)$
3. $y = \frac{1}{2}(x+2)^2$

Example

Use the vertex, axis of symmetry, and y-intercept to graph

$$y = -2(x+1)^2 + 4.$$

Example

Determine the coordinates of the vertex of

$$y = 2x^2 - 8x + 1.$$

IV - Graphs of quadratic functions

a- Sketch by completion of the square

If we wish to find the vertex of a quadratic given in general form

$$y = ax^2 + bx + c,$$

then one approach is to convert it to the form

$$y = a(x - h)^2 + k,$$

where we can read off the coordinates of the vertex (h, k). One way to do this is to **complete the square.**

Example

Write

$$y = x^2 + 4x + 3$$

in the form

$$y = (x - h)^2 + k$$

by *completing the square*. Hence sketch

$$y = x^2 + 4x + 3,$$

stating the coordinates of the vertex.

b- Discriminant

The discriminant of the quadratic equation

$$ax^2 + bx + c = 0$$

is

$$\Delta = b^2 - 4ac.$$

Property

For a quadratic function $y = ax^2 + bx + c$, we consider the discriminant $\Delta = b^2 - 4ac$.

- If $\Delta < 0$, the graph does not cut the x-axis.
- If $\Delta = 0$, the graph *touches* the x-axis.
- If $\Delta > 0$, the graph *cuts* the x-axis twice.

Property

Positive definite quadratics are quadratics which are positive for all values of x. So,

$$ax^2 + bx + c > 0 \quad \text{for all } x \in \mathbb{R}.$$

Test: A quadratic is **positive definite** if and only if $a > 0$ and $\Delta < 0$.

Negative definite quadratics are quadratics which are negative for all values of x. So,

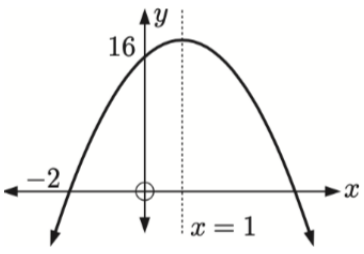
$$ax^2 + bx + c < 0 \quad \text{for all } x \in \mathbb{R}.$$

Test: A quadratic is **negative definite** if and only if $a < 0$ and $\Delta < 0$.

c- Finding equation from a graph

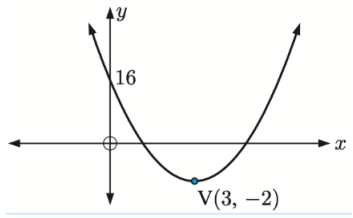
Example

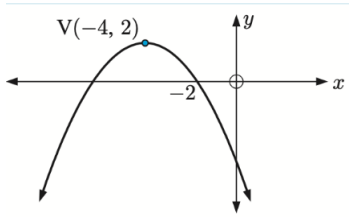
Find the equation of the quadratic function with graph:



Example

Find the equation of each quadratic function given its graph:

1. 

2. 

d- Intersection of graphs

Example

Find the coordinates of the points of intersection of the graphs with equations

$$y = x^2 - x - 18 \quad \text{and} \quad y = x - 3$$

without technology.

e- Problem Solving

Example

A rectangle has length 3 cm longer than its width. Its area is 42 cm^2. Find its width.

Property

For the quadratic function

$$y = ax^2 + bx + c,$$

we have already seen that the vertex has x-coordinate

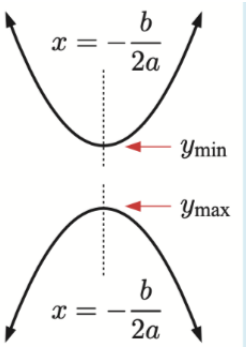
$$x = -\frac{b}{2a}.$$

- If $a > 0$, the **minimum** value of y occurs at

$$x = -\frac{b}{2a}.$$

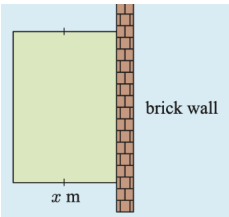
- If $a < 0$, the **maximum** value of y occurs at

$$x = -\frac{b}{2a}.$$



Example

A gardener has 40 m of fencing to enclose a rectangular garden plot, where one side is an existing brick wall. Suppose the two new equal sides are x m long.



1. Show that the area enclosed is given by

$$A = x(40 - 2x) \text{ m}^2.$$

2. Find the dimensions of the garden of maximum area.

V - Sum and product of roots (HL)

Property

If α and β are the roots of the quadratic equation

$$ax^2 + bx + c = 0, \quad a \neq 0,$$

then

$$\alpha + \beta = -\frac{b}{a} \quad \text{and} \quad \alpha\beta = \frac{c}{a}.$$

Proof

Since α and β are the roots, we can write

$$ax^2 + bx + c = a(x - \alpha)(x - \beta).$$

Expanding the right-hand side,

$$a(x - \alpha)(x - \beta) = a\big(x^2 - (\alpha + \beta)x + \alpha\beta\big) = ax^2 - a(\alpha + \beta)x + a\alpha\beta.$$

Comparing coefficients of x and the constant term with $ax^2 + bx + c$,

$$-a(\alpha + \beta) = b \quad \Rightarrow \quad \alpha + \beta = -\frac{b}{a},$$

$$a\alpha\beta = c \quad \Rightarrow \quad \alpha\beta = \frac{c}{a}.$$

Example

The roots of $2x^2 - 7x + 3 = 0$ are α and β. Find, without solving the equation:

1. $\alpha + \beta$
2. $\alpha\beta$
3. $\alpha^2 + \beta^2$
4. $\dfrac{1}{\alpha} + \dfrac{1}{\beta}$

Method. To find a quadratic equation whose roots are α and β, use

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0.$$

Example

The roots of $x^2 - 5x + 2 = 0$ are α and β. Find a quadratic equation with roots $\alpha + 2$ and $\beta + 2$.

Remark. The properties $\alpha + \beta = -\frac{b}{a}$ and $\alpha\beta = \frac{c}{a}$ still hold when the roots α and β are complex.